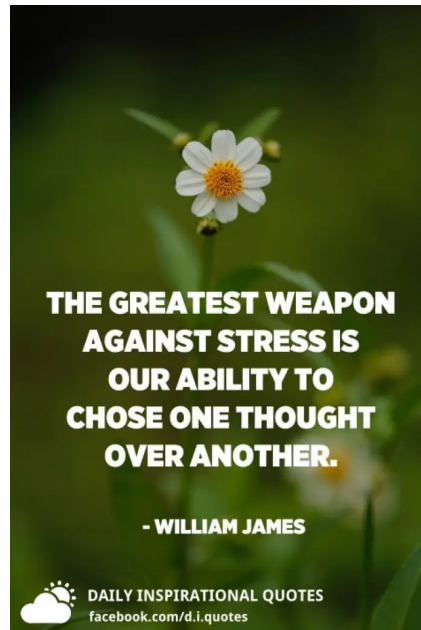




Good

Morning

Content

- Combination basic & properties
- Pascal Triangle
- GCD basics
- special property of GCD
- Delete one

Combination $\rightarrow$ {selection} $\rightarrow$ No. of ways to select something

$(i, j) = (j, i) \rightarrow$ order of selection doesn't matter

Q Given 4 players, count no. of ways of selecting 3 players

{ P_1 P_2 P_3 P_4 }

P_1	P_2	P_3	} <u>4 ways</u>
P_1	P_2	P_4	
P_1	P_3	P_4	
P_2	P_3	P_4	

Q Given 4 players, write the no. of ways to arrange 3 players in 3 slots

{ P_1 P_2 P_3 P_4 }

<u>P₁</u>	<u>P₂</u>	<u>P₃</u>		<u>P₂</u>	<u>P₃</u>	<u>P₄</u>		<u>P₃</u>	<u>P₄</u>	<u>P₁</u>		<u>P₄</u>	<u>P₁</u>	<u>P₂</u>
<u>P₁</u>	<u>P₃</u>	<u>P₂</u>		<u>P₂</u>	<u>P₄</u>	<u>P₃</u>		<u>P₃</u>	<u>P₁</u>	<u>P₄</u>		<u>P₄</u>	<u>P₂</u>	<u>P₁</u>
<u>P₂</u>	<u>P₁</u>	<u>P₃</u>		<u>P₃</u>	<u>P₂</u>	<u>P₄</u>		<u>P₁</u>	<u>P₃</u>	<u>P₄</u>		<u>P₂</u>	<u>P₁</u>	<u>P₄</u>
<u>P₂</u>	<u>P₃</u>	<u>P₁</u>		<u>P₃</u>	<u>P₄</u>	<u>P₂</u>		<u>P₁</u>	<u>P₄</u>	<u>P₃</u>		<u>P₂</u>	<u>P₄</u>	<u>P₁</u>
<u>P₃</u>	<u>P₁</u>	<u>P₂</u>		<u>P₄</u>	<u>P₂</u>	<u>P₃</u>		<u>P₄</u>	<u>P₁</u>	<u>P₃</u>		<u>P₁</u>	<u>P₂</u>	<u>P₄</u>
<u>P₃</u>	<u>P₂</u>	<u>P₁</u>		<u>P₄</u>	<u>P₃</u>	<u>P₂</u>		<u>P₄</u>	<u>P₃</u>	<u>P₁</u>		<u>P₁</u>	<u>P₄</u>	<u>P₂</u>

P₁
P₂
P₃

P₂
P₃
P₄

P₃
P₄
P₁

P₄
P₁
P₂

Selection

For every selection = 6 arrangement = 3!

No. of selections * No. of arrangements for each selection $\Rightarrow$ Total no. of arrangements

$$x * 6 = 24$$

$$x = \frac{24}{6} = 4$$

* Given n distinct elements, in how many ways we can select r elements

$$// \text{ } n \text{ distinct ele, arrange } r \text{ elements} = \frac{n!}{(n-r)!}$$

$$\text{Arrange } r \text{ elements} \Rightarrow r!$$

$$\text{No. of selections for } r \text{ distinct ele} = 1$$

$$\text{No. of selection} * r! = \frac{n!}{(n-r)!}$$

$$\text{No. of selection} = \frac{n!}{r! * (n-r)!}$$

$${}^nC_r = \frac{n!}{r! (n-r)!}$$

* properties of combination

01. No. of ways to select 0 items from N

$${}^N C_0 \Rightarrow \frac{N!}{0! (N-0)!} \Rightarrow 1$$

02. No. of ways to select N items from N

$${}^N C_N = \frac{N!}{N! (N-N)!} \Rightarrow 1$$

03 No. of ways to select r items from N

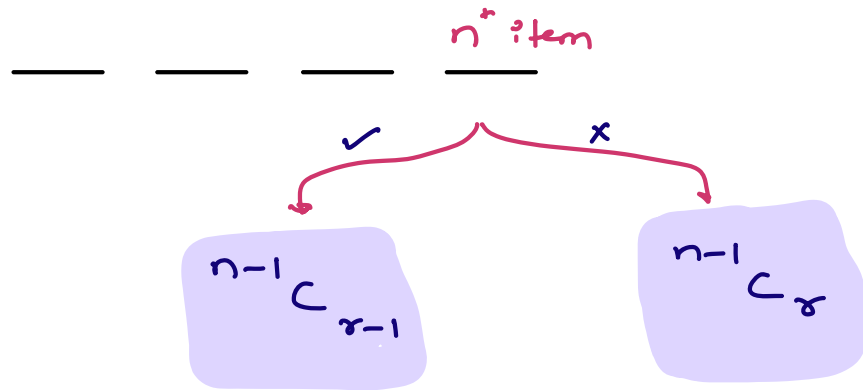
$${}^N C_r = \frac{N!}{r! (n-r)!}$$

04 No. of ways to select $(n-r)$ items from N

$${}^N C_{N-r} = \frac{N!}{(N-r)! (N-(N-r))!} = \frac{N!}{(N-r)! r!}$$

$${}^N C_r = {}^N C_{n-r}$$

* Given n distinct char, select r items



$${}^n C_r = {}^{n-1} C_{r-1} + {}^{n-1} C_r$$

* Pascal Triangle

	1				
→	1	1			
→	1	2	1		
→	1	3	3	1	
	1	4	6	4	1

$$\begin{array}{ccccccccc} {}^0C_0 & & & & & & & & & & \\ & {}^1C_0 & & {}^1C_1 & & & & & & & \\ & & {}^2C_0 & & {}^2C_1 & & {}^2C_2 & & & & \\ & & & {}^3C_0 & & {}^3C_1 & & {}^3C_2 & & {}^3C_3 & \\ & & & & {}^4C_0 & & {}^4C_1 & & {}^4C_2 & & {}^4C_3 & & {}^4C_4 \end{array}$$

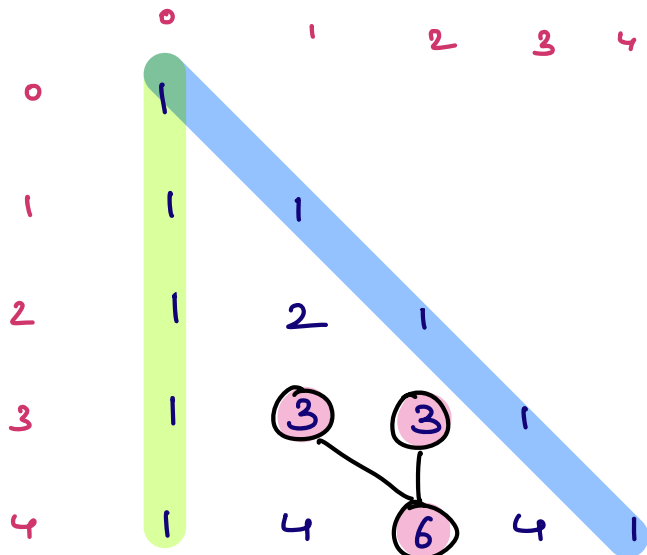
- * **Brute force** → For each & every value (n_{cr}), calculate the factorials & print them.

$$5! = 120$$

$$10! = 3628800$$

$$20! = 2.4 \times 10^{18}$$

factorials grows rapidly & will give overflow



$$* nC_0 = 1$$

$$* nC_n = 1$$

$$* nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$$

$${}^4C_2 = {}^3C_1 + {}^3C_2$$

$$c[i][j] = c[i-1][j-1] + c[i-1][j]$$

* Pseudocode

```
System.out.println("1")
```

```
for (i=1 ; i < n ; i++) {
```

```
    c[i][0] = 1
```

```
    c[i][i] = 1
```

```
    System.out.print ( c[i][0] );
```

```
    for (j=1 ; j < i ; j++) {
```

```
        | c[i][j] = c[i-1][j-1] + c[i-1][j];
```

```
        | System.out.print (c[i][j]);
```

```
    System.out.println ( c[i][i] );
```

```
3
```

TC: $O(n^2)$

SC: $O(n^2)$

8:04 AM → 8:14 AM

GCD $\rightarrow$ Greatest common Divisor

HCF $\rightarrow$ Highest common factor

GCD(A, B) $\rightarrow$ Greatest factor that divides both A & B

GCD(A, B) = x $\rightarrow$ $\left. \begin{array}{l} A \% x = 0 \\ B \% x = 0 \end{array} \right\} x = \text{greatest divisor}$

GCD(15, 25)

1	1
3	5
5	25
15	

Ans = 5

GCD(12, 30)

1	1
2	2
3	3
4	5
6	6
12	10
	15
	30

Ans = 6

GCD(10, -25)

1	-25
2	-5
5	-1
10	1
	5
	25

Ans = 5

GCD(0, 4)

1	1
2	2
3	4
4	
5	
⋮	
⋮	
∞	

Ans = 4

GCD(0, -10)

1	-10
2	-5
3	
4	-2
⋮	-1
10	1
⋮	2
⋮	5
∞	10

Ans = 10

* Properties of GCD

$$01. \text{GCD}(A, B) = \text{GCD}(B, A)$$

$$02. \text{GCD}(A, B) = \text{GCD}(|A|, |B|)$$

$$03. \text{GCD}(0, A) = |A|$$

$$\begin{aligned} 04. \text{GCD}(A, B, C) &= \text{GCD}(A, \text{GCD}(B, C)) \\ &\Rightarrow \text{GCD}(B, \text{GCD}(A, C)) \\ &= \text{GCD}(C, \text{GCD}(A, B)) \end{aligned}$$

* Special property of GCD

$$\text{Given } A, B > 0 \quad \&\& \quad A \geq B \quad \&\& \quad \text{GCD}(A, B) = x$$

$$\begin{cases} A \% x = 0 \end{cases}$$

$$\begin{cases} B \% x = 0 \end{cases}$$

$$\text{GCD}(A - B, B) = x$$

$$(A - B) \% x = 0$$

$$B \% x = 0$$

$$\Rightarrow \underbrace{(A \% x)}_{0} - \underbrace{(B \% x)}_{0} + x) \% x$$

$$\Rightarrow (0 - 0 + x) \% x$$

$$= x \% x = 0$$

$$A, B > 0 \quad \&\& \quad A \geq B$$

$$* \gcd(23, 5) = \overset{A-1B}{\gcd(18, 5)} = \overset{A-2B}{\gcd(13, 5)} = \overset{A-3B}{\gcd(8, 5)}$$
$$\Downarrow$$
$$\underset{\text{A-4B}}{\gcd(3, 5)}$$

$$\text{gcd}(A, B) = \text{gcd}(A - 1B, B)$$

$$= 4CD(A-3B, B)$$

$$= \text{GCD}(A - 2B, B)$$
$$\text{GCD}(A, B) = \text{GCD}(A \% B, B)$$

* Write a function to calculate $\text{GCD}(A, B)$

$$01. \text{GCD}(24, 16) = \text{GCD}(8, 16) = \text{GCD}(8, 16) \rightarrow \text{Infinite loop}$$

✓ $\text{GCD}(A, B) = \text{GCD}(B, A \% B)$

$$* \text{GCD}(24, 16) = \text{GCD}(16, 8) = \text{GCD}(8, 0) \quad \underline{\text{Ans} = 8}$$

$$* \text{GCD}(14, 21) = \text{GCD}(21, 14) = \text{GCD}(14, 7) = \text{GCD}(7, 0) \quad \underline{\underline{\text{Ans} = 7}}$$

$$* \text{GCD}(3, 7) = \text{GCD}(7, 3 \% 7)$$

$$= \text{GCD}(7, 3) \Rightarrow \text{GCD}(3, 7 \% 3)$$

$$\Rightarrow \text{GCD}(3, 1) \Rightarrow \text{GCD}(1, 3 \% 1)$$

$$\Rightarrow \text{GCD}(1, 0) \quad \underline{\underline{\text{Ans} = 1}}$$

$a, b > 0$

```
* int gcd(a, b)
|
|   if (b == 0) return a;
|
|   return gcd(b, a % b);
|
3
```

TC: $O(\log \max(a, b))$

* Given $arr[]$, calculate GCD of entire array

$$arr[3] = \{6, 12, 15\}$$



Ans = 3

```
int gcdarr (int [] A)
{
    ans = A[0]
    for (i=1; i<n; i++) {
        ans = gcd(ans, A[i]);
    }
    return ans;
}
```

TC: $O(n \log(\max \text{ of arr}))$:

* Delete One

Given $arr[N]$ elements, we have to delete one element such that GCD of remaining ele become maximum

$$arr[] = \{ \begin{matrix} 24 & 16 & 18 & 30 & 15 \\ 0 & 1 & 2 & 3 & 4 \end{matrix} \}$$

GCD

24 16 18 30 15	0 1 2 3 4	1	} <u>Ans = 3</u>
24 16 18 30 15		3	
24 16 18 30 15		1	
24 16 18 30 15		1	
24 16 18 30 15		2	

Brute-force $\rightarrow$ Delete an ele & calculate GCD for all the remaining elements while maintaining the max $\rightarrow$ Repeat for all arr elements

TC: $O(n^2 \log \max \dots)$

* Idea 2 Prefix arrays

ans = 0

for (i=0; i<n; i++) {

ans = gcd(ans, arr[i]);

pfgcd[i] = ans;

}

```
int deleteOne (int [] ar)
```

```
    ans = 0;
```

```
    int [] pfgcd : pfgcd[i] = gcd of all ele from  
                    0 to i
```

```
    int [] sfgcd : sfgcd[i] = gcd of all ele from  
                    i to n-1
```

```
    for (i=0; i<n; i++)
```

```
        // delete ar[i] & gcd of ele 0 to i-1 & gcd(i+1, n-1)
```

```
        ar[] = { a0 a1 a2 ... ai-1 ai ai+1 ... an-1 }
```

```
        // left = gcd of ele from 0 to i-1
```

```
        int left = 0
```

TC: $O(n \log \max \text{ of arr })$

```
        if (i > 0) left = pfgcd[i-1];
```

SC: $O(n)$

```
        // right = gcd of ele from i+1 to n-1
```

```
        int right = 0
```

```
        if (i != n-1) right = sfgcd[i+1]
```

```
        int val = gcd(left, right);
```

```
        ans = Math.max(ans, val);
```

```
    }  
    return ans;
```

Dry run



$$arr[] = \{ 24 \quad 16 \quad 18 \quad 30 \quad 15 \}$$

0	1	2	3	4
---	---	---	---	---

$$pfgcd[] = \{ 24 \quad 8 \quad 2 \quad 2 \quad 1 \}$$

0	1	2	3	4
---	---	---	---	---

$$sfgcd[] = \{ 1 \quad 1 \quad 3 \quad 15 \quad 15 \}$$

0	1	2	3	4
---	---	---	---	---

$$left = pfgcd[i-1] = 24$$

$$right = sfgcd[i+1] = 3$$

$$ans = 1$$